Annotated References

Evidential Fusion for Calibrated Edge Traversability on a Wheeled Delivery Robot
Jeff Barner, ECE 532 final project proposal

Cited in the proposal

- Zongbo Han, Changqing Zhang, Huazhu Fu, and Joey Tianyi Zhou. Trusted multi-view classification with dynamic evidential fusion. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(2):2551–2566, 2023. doi: 10.1109/TPAMI.2022.3171983

  Use in this project: Trusted Multi-View Classification (TMC). Each view yields a Dirichlet opinion (belief and uncertainty), and opinions are combined with a reduced Dempster's rule. TMC is the headline baseline and the method our proposed fusion extends with a label-age-discounted map view and accumulation of evidence over the approach. Its conflict term is the quantity checker C2 tests.

- Jing Liang, Dibyendu Das, Daeun Song, Md Nahid Hasan Shuvo, Mohammad Durrani, Karthik Taranath, Ivan Penskiy, Dinesh Manocha, and Xuesu Xiao. GND: Global navigation dataset with multi-modal perception and multi-category traversability in outdoor campus environments. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 2383–2390, 2025. doi: 10.1109/ICRA55743.2025.11128193

  Use in this project: GND, the dataset. It supplies the Jackal robot logs (LiDAR, RGB camera, odometry, SLAM poses) for UMD map 3 and Georgetown, and the human-labeled traversability map whose stairs-or-curb class defines our *hazard* label. CC0 licensed.

- OpenStreetMap contributors. OpenStreetMap. `https://www.openstreetmap.org/copyright`, 2026. Map data © OpenStreetMap contributors, available under the Open Database License (ODbL)

  Use in this project: The map prior. OSM footway tags give the map view's opinion, edit timestamps give label age for the reliability discount, and mapped steps, kerbs and crossings set the risk tiers. Extracts are fetched once as of the GND drive date and committed under the ODbL.

- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mido Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Herve Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL `https://openreview.net/forum?id=a68SUt6zFt`. Featured Certification

  Use in this project: DINOv2. A frozen ViT-S/14 backbone provides the cached camera embedding (the camera view's feature). Using frozen features keeps perception out of scope and the pipeline within the CPU budget.

- Martin A. Fischler and Robert C. Bolles. Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography. *Commun. ACM*, 24(6):381–395, June 1981. ISSN 0001-0782. doi: 10.1145/358669.358692

  Use in this project: RANSAC. Checker C1 fits a ground plane to the raw LiDAR points on an edge with RANSAC and reports the largest step above it, a deterministic geometric check with no learning.

- Michelle Valente, Cyril Joly, and Arnaud de la Fortelle. Fusing laser scanner and stereo camera in evidential grid maps. In *2018 15th International Conference on Control, Automation, Robotics and Vision (ICARCV)*, pages 990–997, 2018. doi: 10.1109/ICARCV.2018.8580635

  Use in this project: Dempster–Shafer fusion of LiDAR and camera in evidential grid maps with hand-set masses. It is the non-learned evidential rung of the fusion ladder and the minimum CPU fallback.

- Qingyang Zhang, Haitao Wu, Changqing Zhang, Qinghua Hu, Huazhu Fu, Joey Tianyi Zhou, and Xi Peng. Provable dynamic fusion for low-quality multimodal data. In *Proceedings of the 40th International Conference on Machine Learning*, ICML'23. JMLR.org, 2023

  Use in this project: Quality-aware Multimodal Fusion (QMF). It weights modalities dynamically by estimated quality, and serves as a second uncertainty-aware fusion baseline beside TMC.

- Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *CoRR*, abs/2107.07511, 2021. doi: 10.48550/ARXIV.2107.07511. URL https://doi.org/10.48550/arXiv.2107.07511

  Use in this project: Conformal prediction tutorial. Per-tier gate thresholds are set by conformal calibration on a held-out split to bound the false-admission rate.

Background, not cited in the proposal body

- Murat Sensoy, Lance Kaplan, and Melih Kandemir. Evidential deep learning to quantify classification uncertainty. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/file/a981f2b708044d6fb4a71a1463242520-Paper.pdf

  Use in this project: Evidential deep learning: a classifier outputs Dirichlet evidence instead of a softmax. It is the single-view basis of TMC and of our per-view evidence heads and loss.

- John W. Lambert and James Hays. Trust, but Verify: Cross-modality fusion for HD map change detection. In *Advances in Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021. URL https://openreview.net/pdf?id=cXCZnLjDm4s

  Use in this project: Map change detection by fusing sensor data with a possibly stale HD map. It is the model for our stale-map injection, where map labels are corrupted on routes with known ground truth.

- Bing Cao, Yinan Xia, Yi Ding, Changqing Zhang, and Qinghua Hu. Predictive dynamic fusion. In *Proceedings of the 41st International Conference on Machine Learning*, ICML'24. JMLR.org, 2024

  Use in this project: Predictive Dynamic Fusion. Related uncertainty-aware dynamic fusion; context for the fusion ladder.

- Xiaoyi Cai, Siddharth Ancha, Lakshay Sharma, Philip R. Osteen, Bernadette Bucher, Stephen Phillips, Jiuguang Wang, Michael Everett, Nicholas Roy, and Jonathan P. How. EVORA: Deep evidential traversability learning for risk-aware off-road autonomy. *IEEE Transactions on Robotics*, 40:3756–3777, 2024. doi: 10.1109/TRO.2024.3431828

  Use in this project: Evidential traversability learning for risk-aware off-road navigation. Closest related work on evidential uncertainty for traversability, applied to a single learned model rather than to fusing sensor and map views.

- Xiaoyi Cai, James Queeney, Tong Xu, Aniket Datar, Chenhui Pan, Max Miller, Ashton Flather, Philip R. Osteen, Nicholas Roy, Xuesu Xiao, and Jonathan P. How. PIETRA: Physics-informed evidential learning for traversing out-of-distribution terrain. *IEEE Robotics and Automation Letters*, 10(3):2359–2366, 2025. doi: 10.1109/LRA.2025.3527285

  Use in this project: Physics-informed evidential learning for out-of-distribution terrain. Related work on making evidential traversability estimates hold up under distribution shift.

- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning - Volume 70*, ICML'17, pages 1321–1330. JMLR.org, 2017

  Use in this project: Temperature scaling. Used to calibrate the softmax fusion baselines so they are compared fairly against the evidential methods.